In NHANES, what predicts missingness in the lipid panel (LDL, triglycerides, HDL, total cholesterol), and which auxiliary variables are recommended for multiple imputation of ffasting-subsample-driven missingness?

Auxiliary variables and missingness in NHANES lipids: what is known

The papers retrieved provide general guidance on auxiliary-variable choice and NHANES-style missingness, but none directly analyze predictors of missingness in the NHANES lipid panel or give a ready-made auxiliary-variable set for fasting-subsample lipids. The answer therefore focuses on principles and related NHANES work that can guide your own variable selection.

Missingness patterns in NHANES-like data

- In NHANES, subsample designs (e.g., fasting subsamples) create **block-like patterns of missingness** rather than item-by-item gaps, and these blocks are associated with survival and other outcomes, implying non-ignorable missingness if ignored (Frölich et al., 2024; Pridham et al., 2022; Chauhan et al., 2023).
- Analyses of NHANES and similar surveys show that missingness in health variables can depend on demographics, health behaviors, and clinical indicators (e.g., BMI, smoking, exercise, comorbidities), which are used as predictors in imputation models (Pan & Chen, 2023; Pridham et al., 2022; Blazek et al., 2020).

Typical predictors used in NHANES imputation case studies

Variable type	Examples used as predictors	Citations
Demographics	Age, sex, race/ethnicity, education, income, marital status	(Pan & Chen, 2023; Blazek et al., 2020)
Health status / comorbidity	General health, hypertension, kidney disease, cardiovascular disease	(Pan & Chen, 2023; Blazek et al., 2020)
Behaviors	Smoking, alcohol, physical activity, diet quality	(Pan & Chen, 2023; Chauhan et al., 2023)
Anthropometrics / labs	BMI, blood pressure, hsCRP, “overweight” status	(Pan & Chen, 2023; Chauhan et al., 2023)

FIGURE 1 Common NHANES-style predictors used in imputation models

How to choose auxiliary variables for lipid-panel MI

Prioritize variables that predict the lipid values, not just missingness

- Theory and simulations show that including auxiliary variables that **only predict missingness but not the lipid itself** can **increase bias** when data are MNAR, especially when missingness is driven by the outcome or both outcome and exposure (Curnow et al., 2024; Mainzer et al., 2024; Curnow et al., 2023).
- Recommended strategy: include **variables most predictive of the partially observed variable** (here LDL, HDL, TG, total cholesterol) plus all variables in the analysis model (Curnow et al., 2024; Mainzer et al., 2024; Madley-Dowd et al., 2023; Hardt et al., 2012).

Avoid problematic auxiliary variables

- Too many weakly correlated auxiliaries, especially in small samples, can produce **downward bias and reduced precision**; a rough rule-of-thumb is not to let the variables-to-complete-cases ratio fall below about 1:3 (Hardt et al., 2012).
- Auxiliary variables that act as **colliders** (share a cause with both lipid level and its missingness) may induce bias and higher SEs and should be excluded where plausible (Curnow et al., 2023).

Role of missingness indicators and advanced MI strategies

- Including **missingness indicators** in MI models can sometimes reduce bias when missingness is informative and MNAR, particularly when missingness depends on the variable itself, but can also increase bias with unmeasured confounding (Sperrin & Martin, 2020; Mainzer et al., 2023; Li et al., 2024).
- NHANES-based and other public-health simulations show that flexible imputation engines (e.g., MICE with CART and carefully chosen auxiliaries) can handle complex block missingness and improve prediction and bias over default models (Pridham et al., 2022; Chauhan et al., 2023).

Conclusion

No paper directly lists predictors of fasting-subsample missingness for NHANES lipids or a canonical auxiliary-variable set. Evidence supports: (1) modeling block missingness using rich demographic, behavioral, and clinical predictors; (2) prioritizing variables that strongly predict the lipid measures themselves; and (3) avoiding variables that only predict missingness, are colliders, or are numerous but weakly related, to limit bias and instability in multiple imputation.

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